

Trained

Place labels decide what counts



train



Large labelled
training set

Corpus-specialized
features

Standard
Strong within-domain
discrimination

Cross-env.
Can inherit training-
domain bias

Training-free, prior work

Added criteria decide what counts

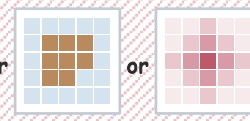


Frozen patch
features

Standard
Selection $\neq$ place
discrimination



Codebook



Segmentation CLS attention

Prior-selected / aggregated features

Cross-env.
Selection criteria may
not transfer

Ours

Let patches decide what counts



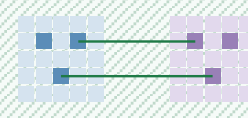
Frozen patch
features

Standard
Match distinctive local
evidence



Within-image

Balanced, matchable patch evidence

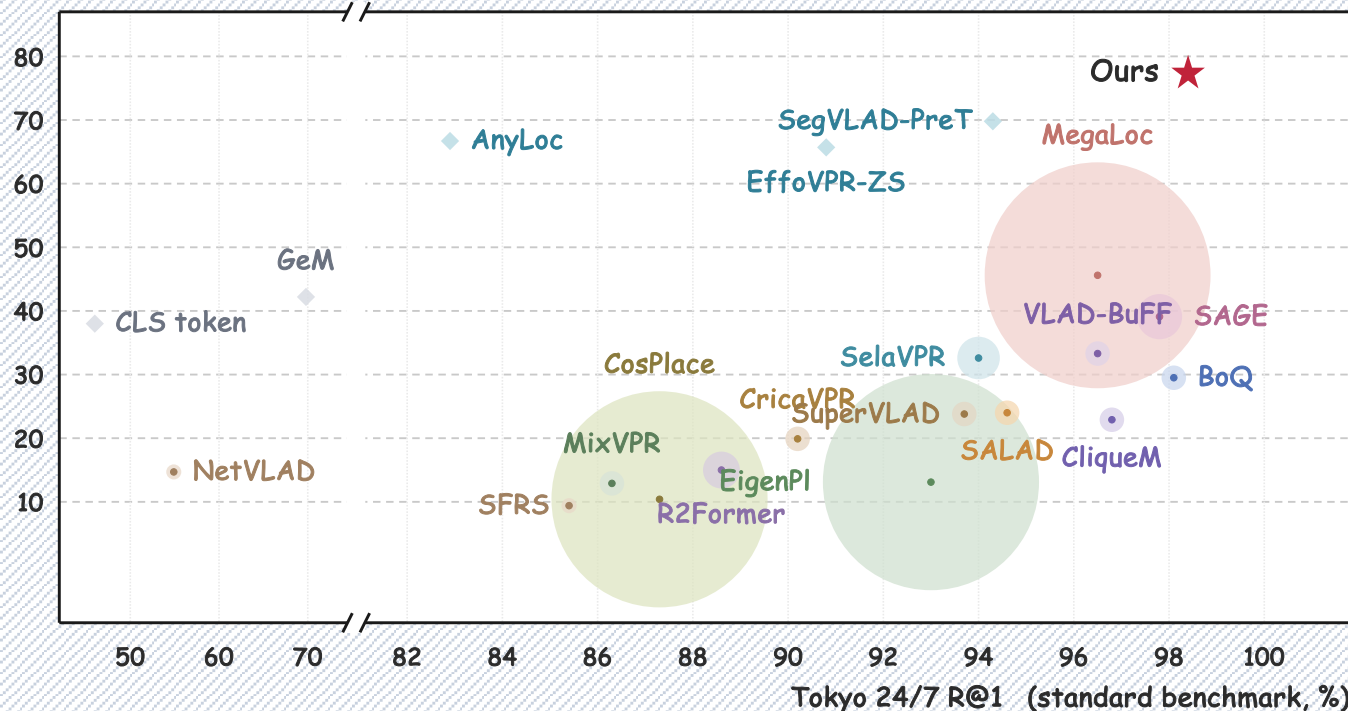


Across-image

Cross-env.
Strong cross-
environment retrieval

Supervision cost vs. cross-environment recall

VP-Air R@1 (cross-environment benchmark, %)



★ ours (training-free) ♦ training-free ● trained ● 1 million images ● 5 million images

Recall across 17 benchmarks

